

Smart Choice Online Voting System

Unit & Integration Testing Report

1. Overview

This report covers the **unit tests**, **feature tests**, and **integration tests** implemented for the Smart Choice Online Voting System built using the **Laravel PHP framework**.

The goal of these tests is to ensure:

- Data validation works correctly
 - OTP-based vote confirmation is secure and reliable
 - Roles & permissions behave correctly
 - Elections and voting logic operate as expected
 - Critical modules function without regression
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2. Modules Tested

✓ User Registration Module

- Email/Student ID uniqueness
- Password validations
- Cloudinary image upload mocking
- Success redirection

✓ Authentication Module

- Login with correct credentials
- Login failure states
- Role-based redirection
- Password reset email sent

✓ OTP System Module

- OTP generation
- OTP hashing & expiry
- OTP email sending
- Invalid & expired OTP rejection

✓ Election Creation Module (Officer)

- Create election with valid inputs
- Add positions
- Add candidates
- Assign voters to elections

✓ Voting Module

- Prevent duplicate votes
- Decrypt & store encrypted votes
- Validate ongoing election timeframe
- Submit vote with OTP verification

✓ Results & PDF Generation

- Server-side tally
- PDF report generation
- Secure access to files

✓ Admin Panel

- Approve officer requests
- Delete voters
- Access audit logs

3. Key Test Cases & Results

3.1 Registration Test

Objective: Ensure new voters can register successfully.

Result: ✓ Passed

Highlights:

- Valid input → user successfully created
 - Duplicate email → returns HTTP 422
 - Password mismatch → validation error
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3.2 Login Test

Objective: Validate role-based login.

Result: ✓ Passed

Highlights:

- Invalid password → shown error message
 - Officer “approved=0” cannot login to officer panel
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3.3 OTP Vote Flow Test

Objective: Confirm secure OTP-based vote confirmation.

Result: ✓ Passed

Highlights:

- OTP generated and emailed
 - OTP must be valid, unused, and unexpired
 - Voting fails if trying twice
 - Encrypted payload stored
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3.4 Duplicate Vote Prevention Test

Objective: Ensure voters cannot vote more than once.

Result: ✓ Passed

Highlights:

- Unique DB constraint triggers 409
 - Prevents vote tampering
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3.5 Election Officer Tests

Objective: Validate election management.

Result: ✓ Passed

Highlights:

- Unauthorized users blocked
 - Officer cannot see who voted for whom
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3.6 PDF Result Generation Test

Objective: Validate PDF generation after the election.

Result: ✓ Passed

Highlights:

- PDF generated using DOMPDF
- Accessible only after voting ends